

Developing an AutoML Solution To Detect Errors in Aircraft Maintenance Documentation

BUAN 407 Undergraduate Capstone

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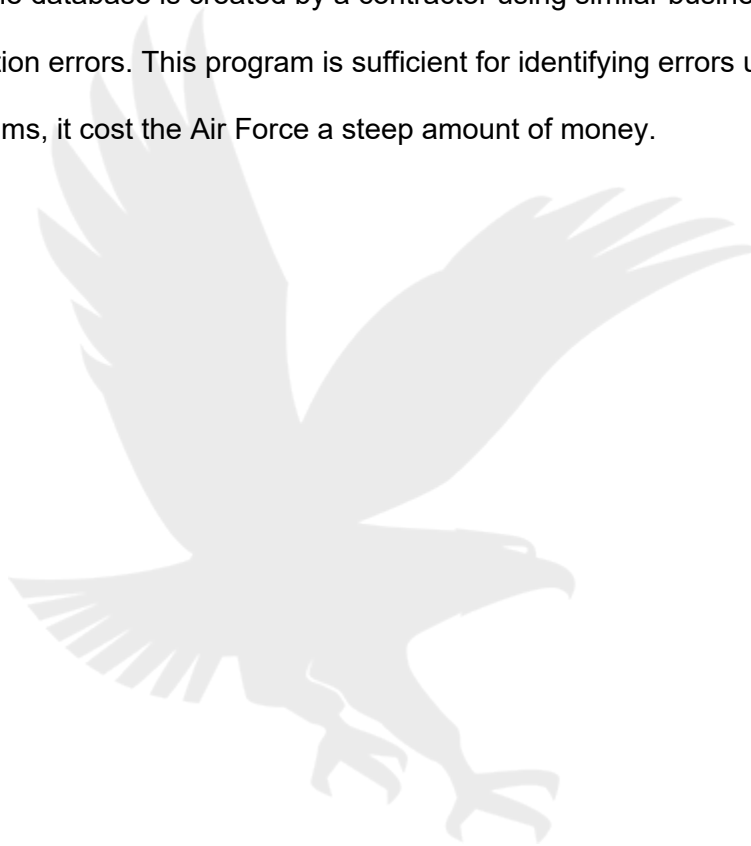
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The integrity and accuracy of data being used in reports and studies utilized by businesses is critical to the decisions being made. On many occasions, inaccurate data had caused great amount of business resources and trust. Data accuracy is being questioned now more than ever with the easy access and availability of public viewable reports such as COVID-19 Dashboards in different countries. Public health policy makers rely on these reports to make decisions so it is very important that data is accurately curated and preserved. A study in the state of Iowa of the importance of data accuracy for policymaking during the pandemic proved that there were challenges in ensuring that data is available and transparent (Srinivas et al., 2021). The study found discrepancies in recorded test results and their respected backdating resulting to lower state COVID positive test results. The after-correction rate doubled the actual SARS-CoV-2 positive test results (2021). We can only imagine the impact of these errors in the data both in policymaking as well as potential spread of COVID-19.

In aircraft maintenance, accuracy and integrity of data is just as important. Data are commonly collected and utilized by various aircraft maintenance departments to operate the business and make decisions. Logisticians rely on accurate part-numbers in order to correctly re-supply stocks. Engineers utilized maintenance time intervals to address reliability issues for certain aircraft parts. Maintenance managers depend on various key performance indicators (KPI) to measure performance and make decisions. These processes all start with data entries.

Part of aircraft maintenance technician's job is to document the work that was being done on the aircraft. Utilizing both paper-based processes and automated systems, they log each step performed to complete the task. With the varying amount of input in the forms and automated systems, ensuring accurate data entry could pose a challenge. Now most organizations have a system or a process in place to address the mistakes that could happen. Some are more robust than the others.

In the U.S. Air Force Aircraft Maintenance Units, most have a program called Data Integrity Teams (DIT). Data Integrity Team is comprised of expert technicians cross-checking forms with the Management Information System (MIS) being used. These technicians checked for common input errors or codes using the published Technical Order (U.S. Air Force T.O. 00-20-2) for documentation. Some units accomplished this manually and some utilize a locally built Microsoft Access database to cross-check the documents with the MIS. More recently, a similar online version of the database is created by a contractor using similar business logic and rules to flag documentation errors. This program is sufficient for identifying errors using pre-set rules, but like most systems, it cost the Air Force a steep amount of money.



Statement of the Problem

Evident from researches performed by Elakramine et al. (2020) and Kornheef et al. (2021), the best solution to address documentation issues is to mitigate the chances of making the errors early in the data entry. This can be accomplished by updating the system of entry and make it much easier for technicians to input data in the Air Force's management information systems. Additions such as drop-downs, check-boxes, or other form of automated entries can eliminate the vast majority of errors generated by manual inputs. Unfortunately, this is also a great undertaking and could potentially cost more resources for the organization. This is the reason why maintenance units are employing Data Integrity Teams to address the problem.

As mentioned earlier, DIT teams employ maintenance technicians to catch and review documentation errors. Every day, Air Force maintenance data analyst run maintenance history reports and upload them to the database for DIT members to review. This process is tedious and also takes away the maintenance technicians from their primary duty. On average, a typical maintenance unit generates up to 60,000 work entries and 3,000 flagged errors a month (Harris, 2021). By regulation, these entries must be reviewed and the accuracy reported to the leadership. This task can be a full-time job for a handful of maintainers and takes them away from more important tasks such as fixing the airplanes. With limited manning and resources available, the need for a more robust and automated solution is never been more important.

This poses the question whether employing machine learning in identifying and correcting errors a viable solution. Can we develop and employ a classification model to find the errors with better accuracy? How can we integrate this solution in existing processes and established information system? And most importantly, can we accomplish this at minimal cost? These questions are what we are going to explore in this research.

Literature Review

Many automated and online systems have been studied and implemented to address the aircraft maintenance documentation issue. Most revolved around the theme of “switching from paper-based approach to complex systems” (Fatine, et al. 2020). On their research and analyses on the application of system modelling approach to replace hard copies of technical manuals and troubleshooting trees, they found that most aviation organizations still rely on the paper-based process. Nature to maintenance troubleshooting and documentation is to follow checklist procedures. The authors approach is to replace this process with system modelling language (SysML). The advance modelling technique would eliminate the documentation-based procedures and switch to an engineering model that is more precise (2020).

While this research shows an excellent example of automating the documentation process in aviation maintenance, it does not quite fit the solution to the accuracy issues in documentation that we experience daily in the Air Force’s aircraft maintenance units. Switching from technical orders and troubleshooting trees to a model-based approach system would be a gigantic cultural shift for USAF aircraft maintenance. However, the modelling approach does limit the human input to the documentation piece of the puzzle that could return benefits to the accuracy of maintenance documents.

Similar research was conducted by Korneef, Verhagen, & Curran (2021) in their attempt to implement a web-based decision support system to address aircraft dispatch and maintenance. Their research revolved around key factors in the aircraft dispatch and maintenance procedures for a typical narrow-body aircraft in the commercial aviation space. According to Korneef, Verhagen, & Curran, one important issue in aircraft maintenance is the amount of time technicians’ use searching through extensive maintenance manuals. This process can take up as much as 30% of the technicians’ time formulating the appropriate

maintenance task (Korneef et al., 2021). Their solution is to implement a fully automated web-based decision support system. This system is designed to automatically retrieve maintenance information from multiple data source and identify appropriate tasks that technicians would normally research manually. Furthermore, the retrieval is based on the type of fault code that technicians can enter. This process can potentially save technicians time by 98%.

This solution is very robust and is also another step in the right direction in eliminating paper-based processes in the aircraft maintenance. However, the research does not directly analyze the impact of inaccurate documentation of the maintenance performed. The authors' system can also be an issue when internet connectivity is not adequate in the flight line as most Air Force aircraft mechanic experiences while performing their duty.

So far, we have yet to find a real solution to the focus of this research which is addressing the inaccuracies in aircraft documentation. There seem to be limited research on this specific topic. It is just as important to understand why maybe there is not much effort in this area given its importance in data-driven decisions. Maybe systems outside the US Air Force aircraft maintenance are very robust that it mitigates potential errors by implementing data validations at the root of the problem – the data entry. Many data driven organizations also employ data professionals that are experts in cleaning and transforming dirty data which addresses this issue in a more reactive manner – devoting resources to it only when the data is needed for analysis. While this might be an acceptable approach in handling data inaccuracies, this process can prove lengthy for data professionals and can delay the data decision-making process. In a high-tempo environment such as the USAF maintenance units, decisions are made at a fast pace. Often data analysis is utilized after the fact due to the lengthy process of delivering insights to the maintenance managers. This is the reason that now more than ever, we need a more proactive approach in identifying potential documentation errors in a real-time

basis to ensure fast delivery of data analysis to consumers. Current solution to this problem relies on user-built data cleansing process such as the Air Forces AutoDIT program.

Analysis of Evidence

In August of 2002, a program was developed by an Air Force maintenance data analyst named MSgt Ronald P. Rogers. Back then, data integrity checks were made using paper-printed records of maintenance histories. A report in the Integrated Maintenance Database System (IMDS) which is the USAF primary management information system for maintenance is ran and saved as paper documents. These report are distributed to the technicians who then have to review these files manually, highlighting and annotating any errors. As you can see this process is very tedious and lengthy when an average aircraft maintenance unit produces roughly 30,000 maintenance documents per month. A big team of Data Integrity Team members is required to maintain this process and perform the task. The aircraft maintenance's first attempt to automate this tedious effort was accomplished by using Microsoft Access program called AutoDIT (Rogers, 2002).

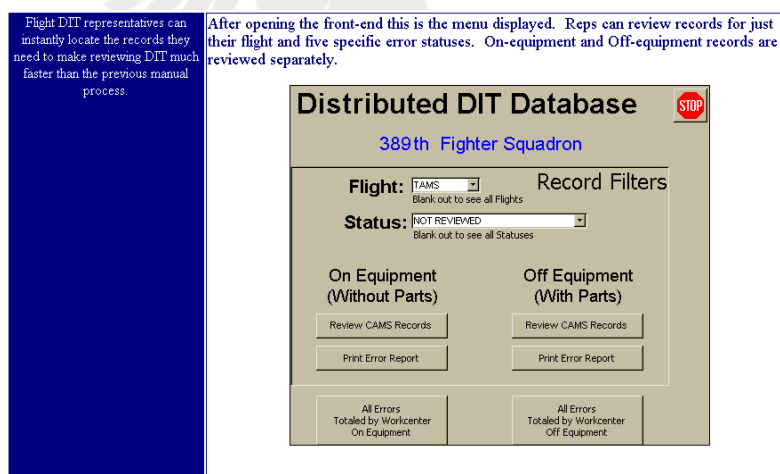


Figure 1. Distributed DIT Database Main Page (Source: Rogers, 2002)

As the name implied, AutoDIT stands for Automated Data Integrity Team by means of an Microsoft Access database. The database is comprised of a backend database where all data imports and error detection process occurs and a frontend database that is distributed to DIT members. The frontend database is what the members utilize to check the errors. The database was later modified to include progress reporting of key performance indicators such as *Squadron Participation Rates, Initial Error Rates, After Correction Rates*, as well common driving factors of data errors.

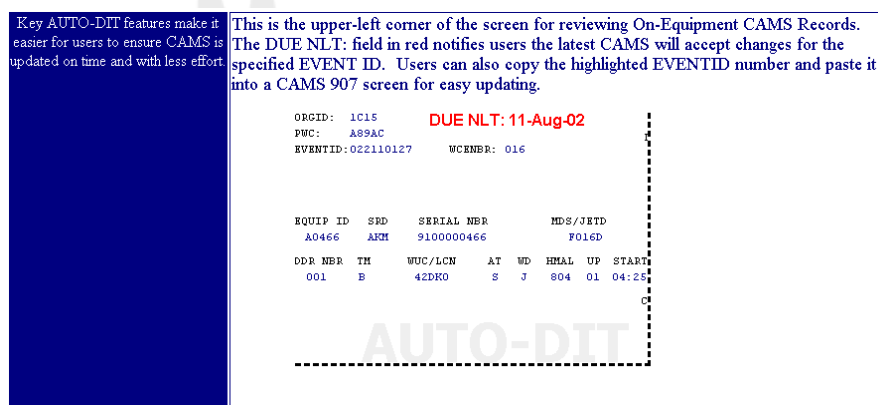


Figure 2. DIT Database Front-End (Source: Rogers, 2002)

BEFORE CORR. ERROR RATE	781	AT	CA	HMAL	OTHER	TM	WUC	NO. ERRORS AFTER CORR.	AFTER CORR. ERROR RATE
7.8%		4	1			19		11	3.6%
#DIV/0!									#DIV/0!
4.2%			2	3		1		4	2.8%
#DIV/0!									#DIV/0!
5.18%	0	4	3	3	0	20	0	15	2.59%
	0%	13%	10%	10%	0%	67%	0%		

These numbers need to be significantly different to show CAMS is being corrected.

Figure 3. DIT Error Drivers (Source: Rogers, 2002)

The DIT database works using hard-coded business rules combining Technical Data references for proper maintenance action code combinations and cumulative knowledge of technicians reviewing the flagged errors. The technicians must review their errors and annotate that corrections were made. The database automates various processes that were previously manually accomplished. For instance, prior to AutoDIT, multiple programs were ran for each maintenance squadron. This results to duplicated efforts that affects system performance of the MIS especially during peak hours. With AutoDIT, the run is centralized and only processed once daily. Manual error identifications were also eliminated by utilizing the AutoDIT database since it is now automated. Moreover, the AutoDIT database eliminated waste of resources by cutting down on printing papers and ink toners.

While this sped up the delivery of the data documents to the consumers and detection of errors, it does have its own drawbacks. Back then in early 2000, Access Database knowledge is not as common as it is now in the Air Force aircraft maintenance units. I had personally experienced as a DIT manager son many occasion when the database would break due to minimal database optimization and server or network issues. This would create downtimes and work stoppage until we can find a database expert to troubleshoot the issue. We have to rely on our expertise to fix problems with the database as there were no AutoDIT helpdesk. More details on this in the next section.

Many iterations of similar process were built and implemented over the years such as AutoDIT versions 2, 3, & 4 which had updated visuals and reporting modules as well as newer technologies incorporating online capabilities such as the Next Generation Auto DIT (NGAT). Fast forward twenty years later, the use of AutoDIT still prevails in many Air Force maintenance units. Aside from the addition of a web-based version of it in a new program called FixDITT (Pavcon LLC, 2021).

According to an e-mail from Mr. Damani, CBM+ Program Office Chief, FixDITT developed by PavCon LLC utilizes free open source technology that makes error flagging easier (Damani, 2021). FixDITT replaced the current NGAT tool in circulation Air Force wide. The new system works similarly with AUtoDIT which relies on user-generated business rules. The error detection works by flagging mismatching maintenance codes and generate a compiled report for the technicians to review. It also included improvements in user management actions making account creations, password resets, and general team management actions more efficient. This process is necessary due to the inherent risk of publishing Air Force data online. This is currently the most widely-used data integrity tool in the aircraft maintenance community. This however, comes at a hefty cost of taxpayer's dollars – the new propose contract is 2M dollars to sustain the program for 5 years.

Limitations of Current Solution

It is first worthy to discuss why the need for accurate historical data is imperative in today's business organization. In a published article by Archana, Zhang, & Churchwell detailing maintenance and inspection as risk factors in helicopter accidents (2019), they emphasized on the integrity and accuracy of maintenance data. They found that maintenance errors and inspection contributed to 14% to 21% of helicopter incidents in the U.S. (Archanam Zhang, & Churchwell, 2019). While the article did not explicitly stated that improper maintenance documentation is a key contributor to helicopter incidents, it could be a symptom of a bad aircraft maintenance culture. If we look at the maintenance culture as whole, it includes proper and accurate documentation. Neglecting this phase of the maintenance process could be an indicator of other problems in the whole maintenance complex. Furthermore, access to and availability of data is much better now more than ever. Data-driven organizations take

advantage of this resources to effectively manage their aircraft fleet and find ways to optimize broken processes. Bad data could lead to bad decisions and cost the organization valuable resources. With this state of mind, that we should put much emphasis and efforts on ensuring maintenance data is curated and verified for accuracy. We can accomplish this with current solutions in AutoDIT and FixDITT but not without limitations.

In the AutoDIT process, there are two key performance indicators that are currently being tracked – *Initial Error Rate (IER)* and *After Correction Rate (ACR)*. The IER measures the errors flagged by the AutoDIT system before the technicians review the document for actual validity. The ACT is the subsequent measure of the errors after the DIT members review the flag errors – based on technical data and technician knowledge. In essence, it is the measure of the number of error that were corrected after conducting reviews. Additionally, the two KPIs can tell us what are the true positive versus the false positive, albeit previous iterations of AutoDIT did not have the function to identify false positives. It was not until recently that it was added in the FixDITT program. These two metrics are calculated using the equations below.

$$IER = \frac{\text{Count of Document with Errors}}{\text{Count of Documents}} * 100$$

$$ACR = \frac{\text{Count of Documents with Errors after Validation}}{\text{Count of Documents}} * 100$$

You may imagine that often, there will be difference between the IER and ACR metrics. In fact, it is a commonly accepted assumption and critical to the reporting process that we track the difference. To illustrate further, we look at Aviano Air Base DIT reports in months of July through November to show an example of the IER and ACR variance. In the table below, the

monthly data was aggregated to give us the average IER compared to the ACR. Then we calculated the difference to show the variance before the initial error rate and after correction rates. The variances were substantial in many maintenance shops ranging from 5% to 30%. The time spent going through the flag errors only to find that they are falsely classified could be substantial.

MX Shop Code	Avg IER	Avg ACR	Variance	Avg False Flag Error Rate
AMETL	5.88	2.02	-3.86	53.41
AFUEL	7.79	5.59	-2.20	43.54
AEGRS	4.86	0.46	-4.41	39.13
ACANN	11.27	5.50	-5.77	38.15
AMXGQ	2.08	0.95	-1.13	38.09
A56CC	17.72	5.62	-12.10	33.19
AELEC	6.08	2.03	-4.05	32.60
APNEU	32.55	4.40	-28.15	32.51
AEWSS	10.53	9.58	-0.95	29.14
AAVTS	8.78	4.43	-4.34	28.53
A56EN	14.65	3.79	-10.86	25.18
AJEIM	0.35	0.35	0.00	24.17
A56FC	12.63	10.12	-2.51	23.20
ASTRU	12.02	5.17	-6.85	23.12

Figure 3. %IER and %ACR Variance and False Flag Error % (Source: Aviano AB, 2022)

Another limitation of the AutoDIT system is the manual adjustment of business rules required to optimize the error identification process. This feature should address the high false flag rates in some maintenance shops, yet as illustrated above the issue persists. It is worthy to point out that to optimize the business rules is easier to accomplish in PavCon's FixDITT, however it still requires both extensive and technical data knowledge. Furthermore, the ability to accomplish this task automatically is very limited.

Lastly, we look at the cost of each solution and present yet another challenge in the sustainment of the program. AutoDIT Access Database is free of charge to any maintenance unit while FixDITT proposed a new sustainment contract for 2M dollars for five years according to the Rapid Sustainment Office (Dosedel, 2022). There are also hidden cost in maintaining the AutoDIT whether it is the amount of manpower or the man-hours involved in maintaining the database. FixDITT absorbs 90% of the database maintenance via contracted Help Desk.

Conclusions

Our brief literature review revealed very little research that was done in our topic of developing and implementing machine learning solution to address documentation errors in aircraft maintenance. There were many related research addressing automation of manual documentation processes which could in-turn increase the accuracy of documentation by limiting human inputs, but I was unsuccessful to find relevant research regarding our topic. Moreover, the use of machine learning is not an uncommon approach in addressing manual processes in the commercial aviation. However, we have yet to see it applied in USAF aircraft maintenance documentation process.

In US Air Force aircraft maintenance units error validation is accomplished by utilizing Data Integrity Teams. This team utilizes a Microsoft Access Database program that automates some of the error capture functions paired with human formal reviews in the past 20 years. Most recent iteration of the AutoDIT program provides many quality of life improvement features and also some form of capturing false flags. However, both systems have their limitations and varying costs to sustain and operate. It is with this conclusion that I present an alternative process utilizing a readily available machine learning algorithmn to either replace or supplement the current data validation efforts.

Recommendations

The main objective of this research is to determine whether it is a viable alternative to use machine learning in identifying documentation errors in aircraft maintenance. We have found that very little research is being done on this topic both in the commercial and within the U.S. Air Force. With this findings, I am compelled to develop a novel solution utilizing readily available platform. Additionally, I will approach this problem set as a business user and not as a machine learning developer or programmer. Needless to say, there will be minimal coding in this solution recommendation.

The first objective of this recommendation is to find a feasible platform that fits these categories; an available system with machine learning capabilities, sustainable for years to come, easily useable, and minimal cost to the organization. The need for an out-of-the-box machine learning solution is critical to the success of this model because aircraft maintenance units do not employ data scientists to create and implement models. They employ data analysts and business users that have limited knowledge of machine learning, therefore it is imperative that we are not overly technical with this solution and select a software that business users and data analyst can grasp without extensive training. This leads us with the options of using Tableau, SAS, or Microsoft Power BI. Each software boasts *AutoML* solutions that fits our needs and varying costs but perform similarly. Currently Power BI is more available to our users so we will go with this platform.

What is AutoML?

In simple terms, AutoML is a pre-packaged analytics software designed to use and implement machine learning solutions with ease. It aims to deploy machine learning models with very little coding and manual model optimization process like *hyper-parameter tuning* (Singla, 2020). It also simplifies the data preparation, model selection, and model comparison tasks that

are common in machine learning deployment pipelines (2020). It allows deploying machine learning solutions in an organization with a variety of guardrails for analysts or business users. Microsoft packaged its AutoML with its analytics Software as Service (SaaS) called Power BI via *dataflow*. This allows for importing data, prepare the data for analysis, and use the data for machine learning utilizing pre-built models (Microsoft, 2021). A caveat, it is only available within Power BI Premium service. Below illustrates the workflow in deploying an AutoML in Power BI.

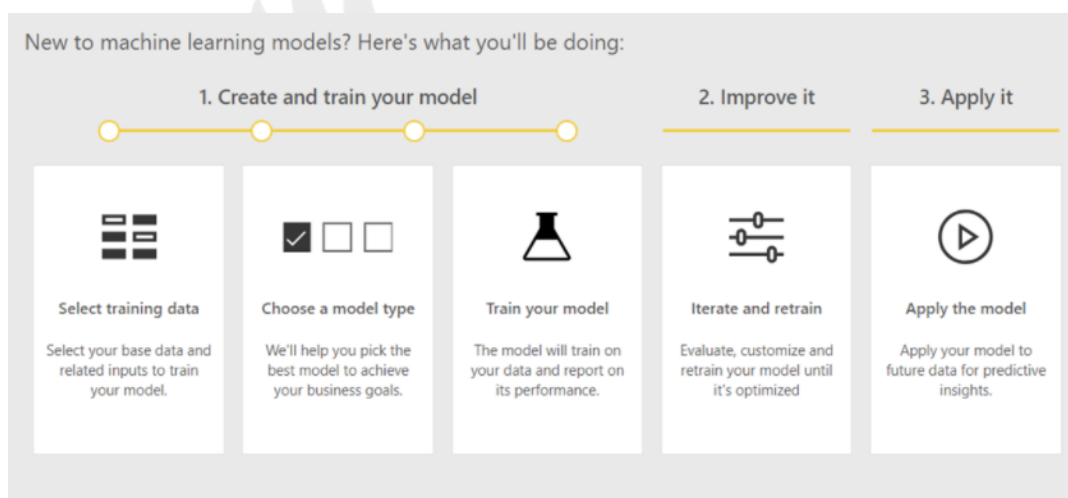


Figure 4. Automated Machine Learning in Power BI (Source: Microsoft, 2021)

The classification AutoML in Power BI returned promising results with the Area Under the ROC (AUC) curve score of 97%. This means that the model can distinguish between errors and not errors at a very high degree. Furthermore, the accuracy of predictions were also highly acceptable at 86% precision. With further model optimization and with better data, the model could provide higher predictive accuracy. Bottom-line, it is very possible to develop and implement a machine learning solution for the Data Integrity process. This AutoML model could address some of the limitation identified in the current solution (AutoDIT and FixDITT).

With a machine learning model, we can mitigate the false flag rate given the higher precision of the AutoML. This classification model recorded as low as 2% false positive rate ($n = 20,000$). Comparatively, FixDITT sample recoded upwards of 25% false flag rate (Fig. 3). It can also address the manual optimization of business rules present in AutoDIT by retraining the model with better and more accurate data. Lastly the cost of utilizing the AutoML capability in Power BI is much a much cheaper alternative than the current sustainment cost of FixDITT (\$60K license cost vs. \$400K FixDITT bill per year).

It is worth noting the “hidden” cost transitioning to a new process and system when implementing the AutoML solution may even more expensive in the beginning. During a discussion with Chief Master Sergeant Joshua Haubold, Aircraft Maintenance Superintendent and Chief of Aircraft Maintenance Programs and Policies in Ramstein Air Base, he explained that “retraining the whole maintenance complex with new solution might prove costly in addition to licensing costs” (2022). Moreover, FixDITT is a “fire and forget” solution, meaning maintenance and training is inclusive in the current contract (Haubold, 2022). Haubold brought up valid points and it does may prove costly to train on a new system. However, AutoDIT is not the only business process that machine learning in Power BI could address. There are many other maintenance processes that calls for new technology and methodology. This research offers the aircraft maintenance domain in the Air Force huge possibilities in pursuing machine learning applications. With analytics tool such as Power BI, we empower our business users to explore possible solutions to both small and big problems.

Additional Information – Automated Machine Learning Methodology

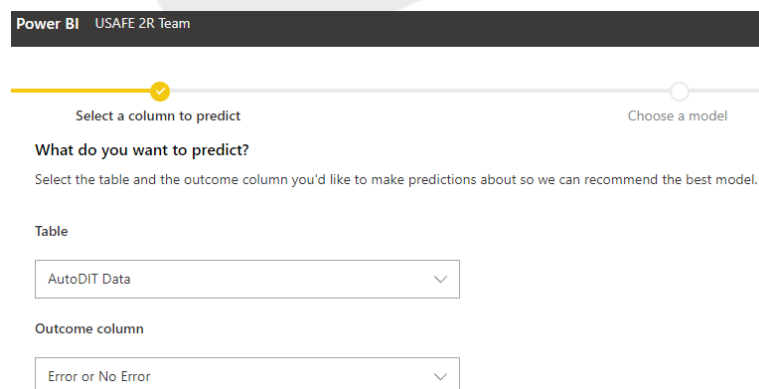
For this machine learning project, we will be using historical data from Aviano Air Base's F-16 C/D model aircraft. We have roughly 2 years of historical information from the AutoDIT database. In essence, we will study the previous error data to develop a classification model using Microsoft Power BI. The sample dataset had 318,666 rows and 18 columns comprised of textual data. The complete data dictionary can be found in Figure 5.

Column Name	Data type	Description	Example
KEY	String	Automatic index of the job document utilizing Equip-ID, Job Control Number, Workcenter Event Number, and Data Document Number	A123420201351234001001
DDRNBR	String	Number designator for data document number	001
DISCREP	String	Narrative of the main work order	Main landing gear inoperable
WCENARR	String	Narrative of each workcenter events	Main landing gear tire requires removal
EQUIPID	String	Identification of equipment	A1234
MDS	String	Mission Design Series	F-16 C/D
TM	String	Type of maintenance	B
WUC	String	Work unit code	13000
AT	String	Action taken by technician	R
WD	String	When discovered code	D
HMAL	String	How malfunction	F
CS	String	Crew size or number of technicians	2
USER	String	ID of user inputting the documentation	ED001C
CORRACTION	String	Narrative of the work accomplished	Removed main landing gear tire
EVTWCEDDR	String	Concatenated value of EVENT ID, WCENUM, and DDR Number	20201351234001001
AFSC	String	Job code of technician working	2A051

ERROR OR NO ERROR	Int	Binary indicator of error or no error	1
CALDATE	Date	Date	Jan 1, 2020

Figure 5. Data Dictionary (Source: Aviano AB, 2022)

The data transformation process was handled in-conjunction with the data ingestion in the *dataflow*. The data required minimal cleaning aside from handling *null* values which were identified and marked as such – for example, null values in the AT column were replaced with “No AT”. There were roughly less than 200 null values in total. We could have also eliminated this null values completely but there is importance in missing values for this project as they denote errors in the documentation. After the data ingestion and transformation process, the data is automatically analyzed by the AutoML engine to determine the best predictors, type of model to use, and the amount of time to train. The next figures illustrate these steps accordingly.



Power BI USAF 2R Team

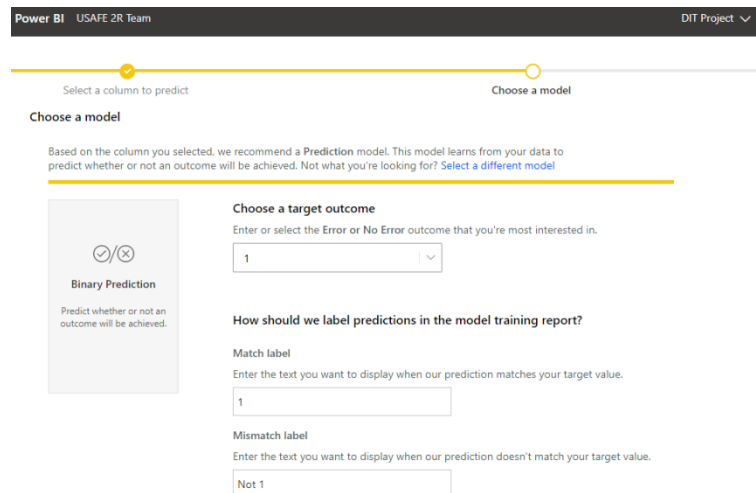
Select a column to predict Choose a model

What do you want to predict?
Select the table and the outcome column you'd like to make predictions about so we can recommend the best model.

Table
AutoDIT Data

Outcome column
Error or No Error

Figure 6. Output selection



Power BI USAF 2R Team DIT Project

Select a column to predict Choose a model

Choose a model

Based on the column you selected, we recommend a Prediction model. This model learns from your data to predict whether or not an outcome will be achieved. Not what you're looking for? [Select a different model](#)

Choose a target outcome
Enter or select the Error or No Error outcome that you're most interested in.

1

How should we label predictions in the model training report?

Match label
Enter the text you want to display when our prediction matches your target value.

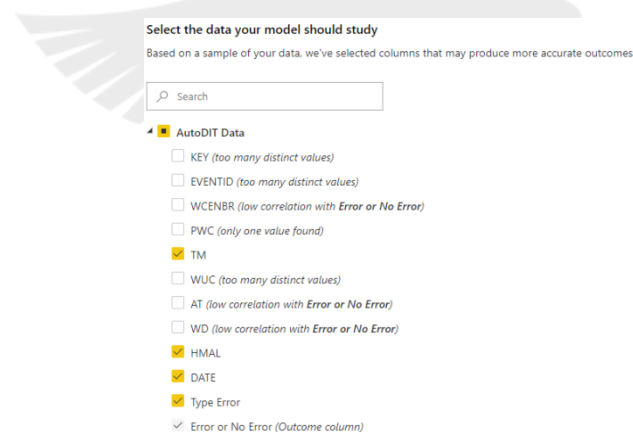
1

Mismatch label
Enter the text you want to display when our prediction doesn't match your target value.

Not 1

Figure 7. Model selection and labeling

The next step is the input variable selection. In Power BI, input variable selection is also selected automatically based on the degree of correlations. Variables are identified with either *low* or *high* correlation and suggests which variables should be used to predict the target variable. The automated variable selection can also be overridden by simply selecting the other variables the analyst deemed important based on domain expertise.



Select the data your model should study
Based on a sample of your data, we've selected columns that may produce more accurate outcomes.

Search

AutoDIT Data

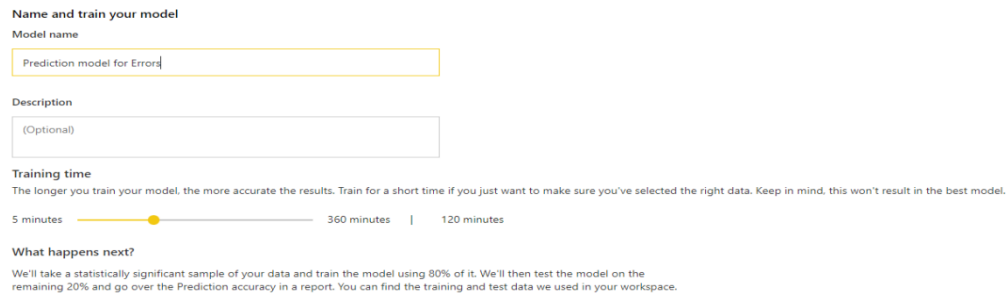
- ☐ KEY (too many distinct values)
- ☐ EVENTID (too many distinct values)
- ☐ WCENBR (low correlation with Error or No Error)
- ☐ PWC (only one value found)
- ☒ TM
- ☐ WUC (too many distinct values)
- ☐ AT (low correlation with Error or No Error)
- ☐ WD (low correlation with Error or No Error)
- ☒ HMAL
- ☒ DATE
- ☒ Type Error
- ☒ Error or No Error (Outcome column)

Figure 8. Input variable selection

The last step is to choose the amount of time desired to train the model. The time ranges from 5 minutes to 360 minutes, with the accuracy directly impacted by the length of training.

Lastly, the AutoML process also handles splitting the data into *train* and *test* sets (80/20 split).

The AutoML model handles necessary processes such as hyper-parameter tuning from here on out and produces a model performance report.



Name and train your model

Model name

Description

Training time
 The longer you train your model, the more accurate the results. Train for a short time if you just want to make sure you've selected the right data. Keep in mind, this won't result in the best model.

5 minutes ☒ 360 minutes ☐ 120 minutes ☐

What happens next?
 We'll take a statistically significant sample of your data and train the model using 80% of it. We'll then test the model on the remaining 20% and go over the Prediction accuracy in a report. You can find the training and test data we used in your workspace.

Figure 9. Model training

Model Performance

AutoML generates a Model Performance report discussing the model performance, accuracy, and training details. Each tab contains relevant information regarding the predictions and its accuracy. The AutoML model performance is measured using the Area under the ROC curve (AUC). This predictive model measured at 97% which denotes an excellent model. Furthermore, the report includes the *Confusion Matrix* measuring the *precision and recall*. This can be adjusted with the desired *probability threshold*. The right balance between *precision* and *recall* must be observed to ensure the acceptable numbers of false positives and false negatives. For this model, this probability threshold is 0.93.

MODEL PERFORMANCE

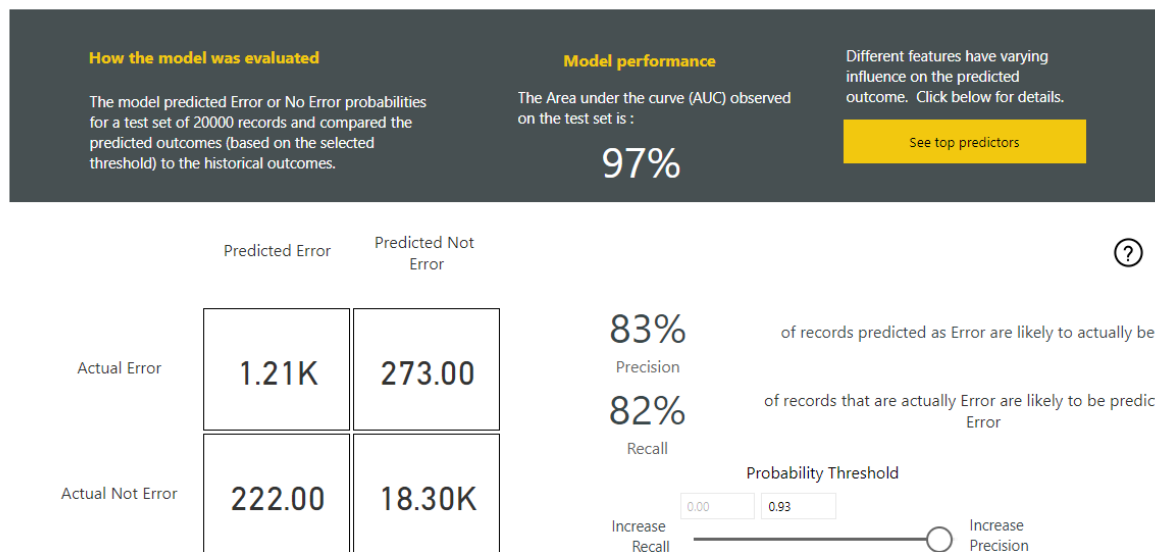


Figure 10. Model Performance

We can also learn more about the top predictors of the classification model by clicking the “See top predictors” button on this report.

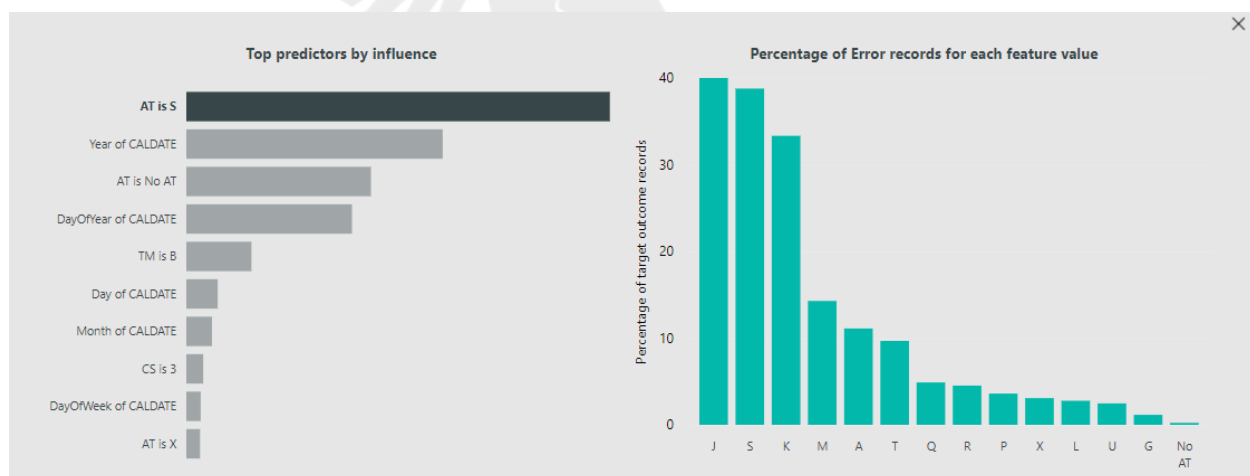


Figure 11. Top predictors by influence

The Accuracy Report explains the accuracy of the model using Cumulative Gains Chart and the ROC Curve. The Gains Chart shows the percentage of rows with the target outcome can be detected by targeting a percentage of total rows. The ROC Curve tells you how capable the model is in distinguishing between the positive outcome and the negative outcome. When reading this charts, we want to be as close as possible to the top left value where the True Positive is at the highest and the False Positive is at the lowest.

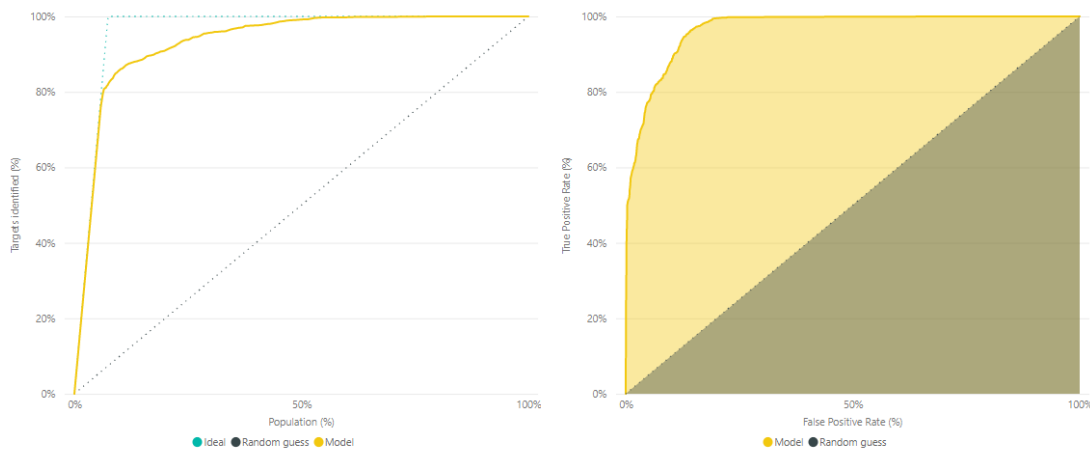


Figure 12. Cumulative Gains Chart and the ROC Curve

Training Details

The last portion of the report discusses the machine learning training process. It highlights the number of iterations and a comparative look of algorithms that were utilized during the training. It also indicates which algorithm performed best. For this model, the algorithm *Pre-fitted soft voting classifier* performed the best.

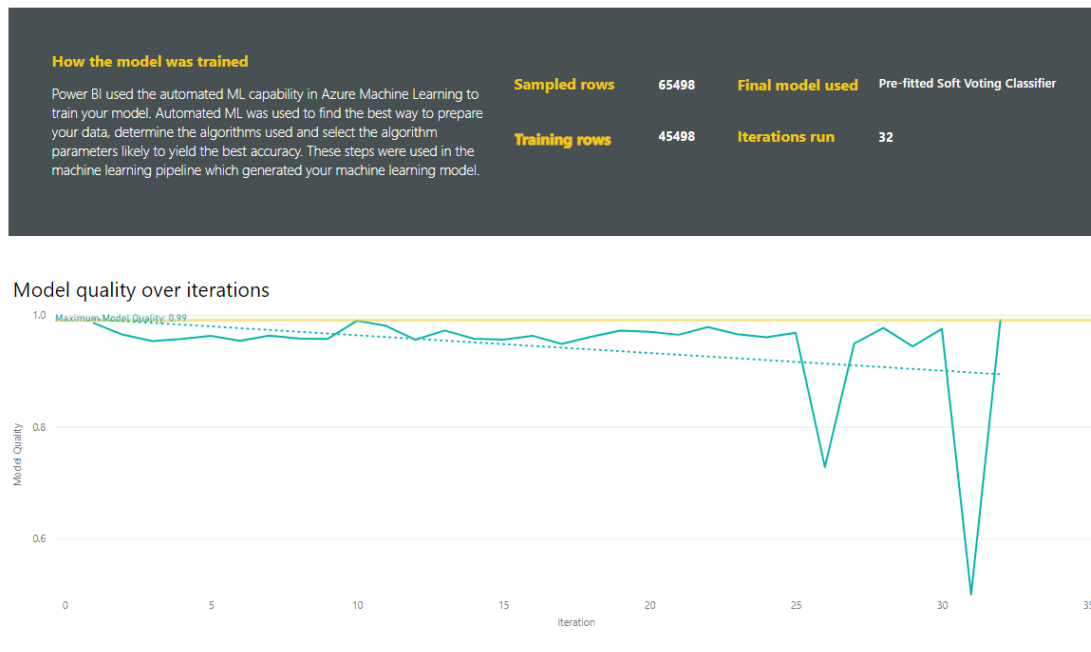


Figure 13. Training details and model quality

Finally, the last graph indicates which model ultimately was used for the prediction. For this project, the AutoML selected an Ensemble machine learning models which utilized combination of Classifiers to predict the outcome values. See Figure 14.

Ensemble machine learning models

Ensemble models use multiple learning algorithms to obtain better predictive performance than may be obtained from a single learning algorithm. Ensemble models are useful for improving accuracy in certain cases.

Automated ML in Power BI generates ensemble models, if they are found to be optimal. If an ensemble model is used, then the constituent model details will be presented below.

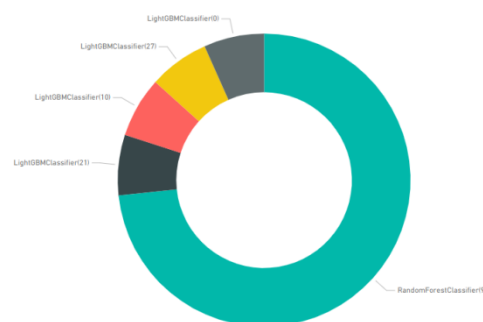


Figure 14. Ensemble machine learning model

Based on the training model, we can predict whether there is an error in a maintenance document with great accuracy. Utilizing AutoML in Power BI is also easy to accomplish given its guard-rails approach in building a model. Applying the model in a live data will generate entities that can be utilized to build a dashboard such as the Data Integrity Dashboard in Figure 15.

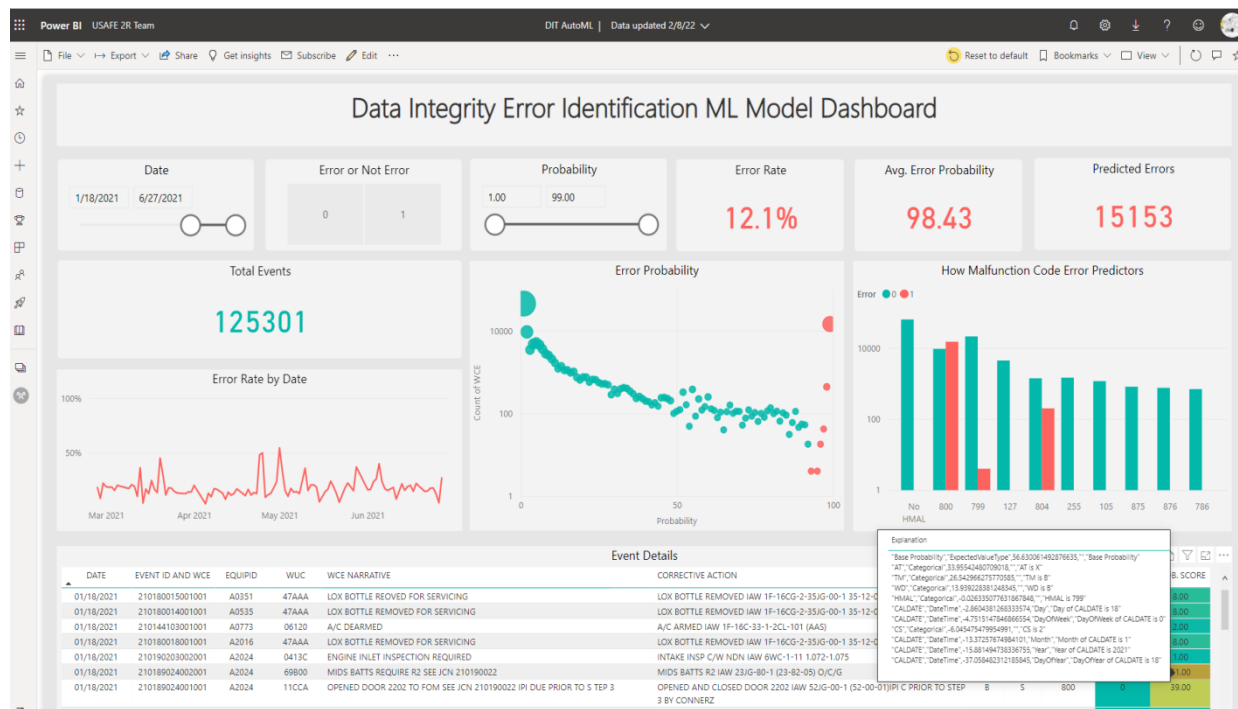


Figure 15. Data Integrity Team Dashboard

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